

Problem 1. Suppose $P \in \mathbb{R}[x]$, P have a number finite of root in $\mathbb{R}^2$ and P is squarefree, demonstrate

Proof. For a polynomial P , we use CAD on it, and then we can obtain a sequence of samples, denoted $x_1 - 1 = x_0, x_1, x_2, \dots, x_s, x_{s+1} = x_s + 1$. In this sequence, $x_1, \dots, x_s$ are the critical indices. We choose one element $\bar{x} \in (x_i, x_{i+1})$, $i = 1, 2, 3, \dots$

According to the properties of CAD, $\forall x \in (x_i, x_{i+1})$, $P(x)$ keep a constant sign, and P is squarefree, so $\forall x \in (x_i, x_{i+1})$, $P(x)$ have the same number of roots. We can arrange the roots in increasing order as $y_1, y_2, \dots, y_t$. $\forall j = 1, 2, \dots, t$, $f(\bar{x}, y_j) = 0$ and $\frac{\partial f}{\partial y}(\bar{x}, y_j) \neq 0$. By the Implicit Function Theorem, we obtain $\forall (\bar{x}, y_j), \exists g_j \in \mathcal{C}^1(I_{\bar{x}})$, so that $\forall a \in I_{\bar{x}}, f(a, g_j(a)) = 0$, supposing that $x_i \notin I_{\bar{x}}$. So we can construct a sequence of functions $h_1, h_2, \dots, h_t$, $h_j(\bar{x}, g_j(x)) = f(\bar{x}, g_j(x)) = 0$

If $\exists a, b = 1, 2, 3, \dots, t, a \neq b$, h_a, h_b intersect, the intersection point lies in the sequence $(x_i)_i$, which contradicts our hypothesis. So in the interval I_x , $(h_j)_j$ don't intersect.

Using Cauchy's bound, $\forall h_j, \forall a \in I_{\bar{x}}$, a fixed, the root b of h_j satisfies $|b| < +\infty$. Now we can construct a sequence of dots $c_0, c_1, \dots, c-m \in (x_i, x_{i+1})$, and their neighborhood, which satisfy $\forall k, l, I_{c_k} \cap I_{c_l} \neq \emptyset$ and $\bigcup_{i=0}^m I_{c_i} = (x_i, x_{i+1})$. For $\bar{x} \in I_{c_k} \cap I_{c_{k+1}}$, $h_{c_k}(\bar{x}, g_{c_k}(\bar{x})) = h_{c_{k+1}}(\bar{x}, g_{c_{k+1}}(\bar{x})) = 0$. In conclusion, by Unioning all the neighborhood, We can obtain a function $h_{i,j} : (x_i, x_{i+1}) \rightarrow \mathbb{R}$. $\square$